EDUARDO DUEÑEZ

Associate Professor of Mathematics at The University of Texas at San Antonio

@ eduardo.duenez@utsa.edu | San Antonio, TX, USA | www.researchgate.net/profile/Eduardo-Duenez | eduenez

EDUCATION

Ph.D. in Mathematics
Princeton University

2001 | Thesis advisor: Peter Sarnak.

B.Sc. in Mathematics
University of Guanajuato, Mexico

1996

PROFESSIONAL EMPLOYMENT

Research Consultant
Gibran.ai (AI startup; seed funding from Together Fund)

2025 – 2026 | Full-time research consulting, June-December 2025.

Associate Professor
The University of Texas at San Antonio

2011 – Present

Assistant Professor
The University of Texas at San Antonio

2005 – 2011

Presidential Fellow
The University of Texas at San Antonio

2004 – 2005

J. J. Sylvester Assistant Professor
The Johns Hopkins University

2002 – 2004

Research Fellow
American Institute of Mathematics, Palo Alto

2001

RESEARCH INTERESTS

My research revolves around analysis in a broad sense. I have worked on random matrix theory (Dyson's "Threefold Way"; linear group ensembles) and number theory (zeros of L-functions); nowadays, my research is on the foundations of analysis and probability, as well as both theoretical and applied aspects of real-valued computing.

REFEREED PUBLICATIONS

1. E. Dueñez, A. S. Hamakiotes, S. J. Miller. *Sums of Powers by L'Hospital's Rule.* Fibonacci Quart. **63**:3 (2025) 603–613.
2. Xavier Caicedo, Eduardo Dueñez, José N. Iovino. *Metastable convergence and logical compactness.* Beyond First Order Model Theory, Volume II, CRC Press, Boca Raton, FL (2023) 3–41.

RESEARCH AWARDS & GRANTS

- **NSF Grant: "Model Theory and Ergodic Theorems" (2015 – 2019)**
  Co-PI with José N. Iovino.
- **AMS/Tensor-SUMMA Grant (2007 – 2008)**
  Supported UTSA's Problem-Solving Seminar.

PREPRINTS & WORK IN PROGRESS

- E. Dueñez, J. Iovino, T. Matos-Wiederhold, L. Salvetti, F. D. Tall. *Complexity of deep computations via topology of function spaces* (In preparation.) arXiv:2601.00528.
- Samson Alva, E. Dueñez, J. N. Iovino, C. Walton. *Approximability of deep computations* (To appear in Mathematical Structures in Computer Science.) arXiv:2409.06064.
- Peter G. Casazza, Eduardo Dueñez, José N. Iovino. *Nondefinability of the spaces of Tsirelson and Schlumprecht* (In press; scheduled to appear in print October 2026.)
- Eduardo Dueñez, José N. Iovino. *Analysis: The Study of Real-Valued and Random-Valued Structures* (Book in preparation.)

PROFESSIONAL/EDITORIAL SERVICE

- Associate Editor, Mathematics Magazine (2016 – 2019).
- Co-editor, Problems and Solutions column, Mathematics Magazine (2016 – 2019).
- Co-organizer (with J. Iovino), session "Topological Methods in Model Theory", Meeting of the AMS Central Section, University of Texas at San Antonio (2024).
- Reviewer of manuscripts for Princeton University Press, Oxford University Press.
- Referee. Annals of Pure and Applied Logic, Bulletin of the London Mathematical Society, Communications

3. Eduardo Dueñez, José N. Iovino. *Model theory and metric convergence II: Averages of unitary polynomial actions.* Contemp. Math. **775** (2021) 85–114.
4. Eduardo Dueñez, José N. Iovino. *Model theory and metric convergence I: Metastability and dominated convergence.* Beyond First Order Model Theory, Volume I, CRC Press, Boca Raton, FL (2017) 131–187.
5. E. Dueñez, D. K. Huynh, J. P. Keating, S. J. Miller, N. C. Snaith. *A Random Matrix Model for Elliptic Curve L-functions of Finite Conductor.* Journal of Physics A: Mathematical and Theoretical **45**:11 (2012) 115207 (32 pp).
6. E. Dueñez, D. K. Huynh, J. P. Keating, S. J. Miller, N. C. Snaith. *The Lowest eigenvalue of Jacobi random matrix ensembles and Painlevé VI.* Journal of Physics A: Mathematical and Theoretical **43**:40 (2010) 405204 (27 pp).
7. E. Dueñez, D. W. Farmer, S. Froehlich, C. P. Hughes, F. Mezzadri, T. Phan. *Roots of the Derivative of the Riemann Zeta Function and of Characteristic Polynomials.* Nonlinearity **23** (2010) 2599–2621.
8. E. Dueñez, S. J. Miller. *The effect of convolving families of L-functions on the underlying group symmetries.* Proc. London Math. Soc. **99**:3 (2009) 787–820.
9. Stephanie Deacon, Eduardo Dueñez, José N. Iovino. *A public-key threshold cryptosystem based on residue rings.* J. Discrete Math. Sci. Cryptogr. **10**:4 (2007) 559–571.
10. E. Dueñez, S. J. Miller. *The low lying zeros of a $\mathrm{GL}(4)$ and a $\mathrm{GL}(6)$ family of L-functions.* Compos. Math. **142**:6 (2006) 1403–1425.
11. E. Dueñez. *"Harder" ensembles of orthogonal matrices.* Experiment. Math. **15**:3 (2006) 273–277.
12. E. Dueñez, S. J. Miller, A. Roy, H. Straubing. *Incomplete quadratic exponential sums in several variables.* J. Number Theory **116**:1 (2006) 168–199.
13. E. Dueñez. *Random matrix ensembles associated to compact symmetric spaces.* Comm. Math. Phys. **244**:1 (2004) 29–61.

in Mathematical Physics, Experimental Mathematics, Journal of Number Theory, Journal of Physics A, Mathematics Magazine, The Ramanujan Journal, Indian Journal of Mathematics, Random Matrices: Theory and Applications, and others.

- NSF panelist; proposal reviewer for CONACyT (Mexico).
- Judge of student abstracts and posters, SACNAS (2010 – 2018).

PROFESSIONAL MEMBERSHIPS

- American Mathematical Society (AMS)
- Mathematical Association of America (MAA)
- Association for Symbolic Logic (ASL)
- Society for Advancement of Chicanos and Native Americans in Science (SACNAS)